Water Pollution

Introduction

Water pollution refers to the contamination of water bodies such as rivers, lakes, oceans, groundwater, and reservoirs by harmful substances that adversely affect the quality of water, human health, and aquatic ecosystems. Pollutants may be physical, chemical, biological, or radioactive substances that alter the natural characteristics of water.

According to the World Health Organization, access to clean water is essential for human survival and sustainable development. However, rapid industrialization, urbanization, and population growth have significantly increased water pollution worldwide.

Definition of Water Pollution

Water pollution can be defined as:

"The undesirable change in the physical, chemical, and biological characteristics of water that makes it harmful to living organisms and unsuitable for intended uses."

Sources of Water Pollution

The sources of water pollution are broadly classified into two categories:

1. **Point Sources**
2. **Non-Point Sources**

1. Point Sources of Water Pollution

Point sources are identifiable and localized sources from which pollutants are discharged directly into water bodies.

Characteristics

- Easily identifiable.
- Pollution originates from a single location.
- Easier to monitor and regulate.

Examples

- Industrial discharge pipes
- Municipal sewage outlets
- Wastewater treatment plants
- Oil refinery effluents
- Power plants

Major Point Sources

A. Industrial Effluents

Industries release large quantities of toxic substances into water bodies.

Pollutants include:

- Heavy metals (lead, mercury, cadmium, chromium)
- Acids and alkalis
- Organic chemicals
- Dyes and solvents
- Petroleum products

Effects:

- Toxicity to aquatic organisms
- Bioaccumulation in food chains

- Reduction of dissolved oxygen
- Human diseases and poisoning

B. Municipal Sewage

Domestic wastewater contains:

- Human excreta
- Food residues
- Soaps and detergents
- Organic matter
- Pathogenic microorganisms

Effects:

- Increase in Biological Oxygen Demand (BOD)
- Oxygen depletion
- Spread of water-borne diseases
- Eutrophication

C. Thermal Power Plants

Power stations discharge hot water into rivers and lakes.

Effects:

- Increase in water temperature
- Decrease in dissolved oxygen
- Disturbance of aquatic ecosystems
- Death of temperature-sensitive species

2. Non-Point Sources of Water Pollution

Non-point sources are diffuse sources of pollution that do not originate from a single identifiable location.

Characteristics

- Widely scattered.
- Difficult to control.
- Pollution is carried by rainfall and runoff.

Examples

- Agricultural runoff
- Urban runoff
- Soil erosion
- Atmospheric deposition

Causes of Water Pollution

The causes of water pollution can be classified into natural and human-induced (anthropogenic) causes.

A. Natural Causes of Water Pollution

Natural processes can also contaminate water.

1. Soil Erosion

Rain and wind carry soil particles into rivers and lakes.

Effects:

- Increased turbidity
- Reduced light penetration
- Destruction of aquatic habitats
- Sedimentation

2. Volcanic Eruptions

Volcanic ash and gases enter water bodies.

Effects:

- Increased acidity
- Elevated mineral content
- Toxic effects on aquatic life

3. Decay of Organic Matter

Decomposition of dead plants and animals releases nutrients.

Effects:

- Increased nutrient concentration
- Reduction of dissolved oxygen
- Growth of microorganisms

4. Natural Disasters

Floods and landslides transport sediments and contaminants into water bodies.

Effects:

- High turbidity
- Contamination by debris and pathogens
- Destruction of aquatic ecosystems

B. Anthropogenic (Human-Induced) Causes of Water Pollution

Human activities are the major contributors to water pollution.

1. Domestic Sewage and Wastewater

Domestic activities generate large amounts of wastewater.

Sources

- Bathrooms
- Kitchens
- Toilets
- Laundry

Pollutants

- Organic matter
- Nutrients (nitrogen and phosphorus)
- Detergents
- Pathogens

Effects

- Oxygen depletion
- Water-borne diseases
- Algal blooms
- Foul smell

2. Industrial Activities

Industries are among the largest sources of water pollution.

Pollutants

- Heavy metals
- Toxic chemicals
- Oils
- Acids
- Radioactive materials

Major Polluting Industries

- Textile industries
- Tanneries
- Chemical industries
- Paper and pulp industries
- Pharmaceutical industries

Effects

- Toxicity
- Cancer and neurological disorders
- Bioaccumulation
- Ecosystem degradation

3. Agricultural Activities

Modern agriculture heavily depends on chemicals.

Agricultural Pollutants

a) Fertilizers

Contain:

- Nitrogen
- Phosphorus
- Potassium

Effects:

- Eutrophication
- Algal blooms
- Oxygen depletion

b) Pesticides

Include:

- Insecticides
- Herbicides
- Fungicides

Effects:

- Toxicity to aquatic organisms
- Biomagnification
- Contamination of groundwater

c) Animal Wastes

Effects:

- High organic load
- Pathogen contamination

- Nutrient enrichment

4. Urbanization and Population Growth

Rapid urban expansion increases wastewater production.

Sources

- Solid waste dumping
- Construction activities
- Sewage discharge

Effects

- Increased sedimentation
- Pathogen contamination
- Chemical pollution

5. Oil Pollution

Oil enters water through:

- Oil spills
- Tanker accidents
- Offshore drilling
- Refinery discharges

Effects

- Destruction of marine ecosystems
- Death of fish and birds
- Reduced oxygen transfer
- Toxic effects on marine life

6. Mining Activities

Mining generates large quantities of waste materials.

Pollutants

- Heavy metals
- Acids
- Suspended solids

Effects

- Acid mine drainage
- Metal contamination
- Destruction of aquatic habitats

7. Plastic Pollution

Plastic waste enters water bodies through improper disposal.

Types

- Plastic bags
- Bottles
- Microplastics
- Packaging materials

Effects

- Ingestion by aquatic organisms
- Entanglement of marine animals
- Release of toxic chemicals

- Food chain contamination

8. Radioactive Pollution

Sources include:

- Nuclear power plants
- Medical facilities
- Research laboratories

Effects

- Genetic mutations
- Cancer
- Long-term ecosystem damage

9. Thermal Pollution

Caused by:

- Power plants
- Industrial cooling systems

Effects

- Reduced dissolved oxygen
- Increased metabolic rates of aquatic organisms
- Alteration of species composition

10. Atmospheric Deposition and Acid Rain

Air pollutants dissolve in rainwater and enter water bodies.

Pollutants

- Sulfur dioxide (SO_2)
- Nitrogen oxides (NO_x)

Effects

- Acidification of lakes and rivers
- Fish mortality
- Damage to aquatic vegetation